THE DRAWING(S) AND THE INFORMATION CONTAINED HEREIN ARE THE PROPERTY OF **WOOJIN INDUSTRIAL SYSTEMS CO., LTD.** THEY SHALL NOT BE DISCLOSED, REPRODUCED OR USED, IN ANY MANNER WITHOUT THE PRIOR WRITTEN CONSENT OF **WOOJIN INDUSTRIAL SYSTEMS CO., LTD.**

SEPTA SCHEDULER
SOFTWARE DESIGN DESCRIPTION

DISTRIBUTION	

Revision Records

Revision	Description	Date
1.0	First Issued	February 17,2011
1.1	4.1.6 Login Interface removed due to SEPTA network already has Log administration.	April 7,2011

Table of Contents

1. Introduction 3
1.1. Purpose 3
1.2. Scope 3
1.3. Acronyms and Abbreviations 3
1.4. References 3
2. Decomposition Description 3
2.1. General Description 4
2.2. Data Decomposition 9
3. Dependency Description 12
3.1. Inter-process dependency - Data flow diagram 12
3.2. Interporcess dependency - Class Diagram 18
4. Interface Description 23
4.1. External Interface 23
5. Detailed Description 28
5.1. Module Detailed Design 28
5.2. Data Detailed Design 37

1. Introduction

1.1. Purpose

This document (Software Design Description, here-in-after called as 'SDD') Description software design for Septa Editor Unit software.
This document shall be used by project Manager, configuration manager, system designer, architect and software developer through all over the software development lifecycle.

1.2. Scope

SEPTA Scheduler Software is to create downloadable update file on vehicle after editing announcement data relevant to schedule by entering or loading the schedule that SEPTA EMU serves.
(Service schedule, Service routes, LED display data, LCD display data and announcement data edit)

1.3. Acronyms and Abbreviations

Acronyms & Abbreviations	Description
PP	Primary Player
SP	Secondary Player
PII	Passenger Information Indicator
FDI	Front Destination Indicator
SDI	Side Destination Indicator

1.4. References

The following documents are referenced

Publisher	Document Number	Title
IEEE	1016-1998	IEEE Recommended Practice for Software Design Description

2. Decomposition Description

This section describes the division of the software into design entities. Each design entity is further divided into a distinct grouping of source code modules at the detailed design level.

2.1. General Description

SEPTA Scheduler Software is able to modify and edit the service data as the software that plays an interface role between manager and devices.

SEPTA Scheduler Software operates in the following environment:

- SEPTA Editor Software is usable on Laptop PC or Desktop equipped with the X86 Processor.
- OS is able to perform on the version higher than the WinXP.
- The database that is used on SEPTA Editor is the MS access.
- Development language is the Visual Studio 2008.

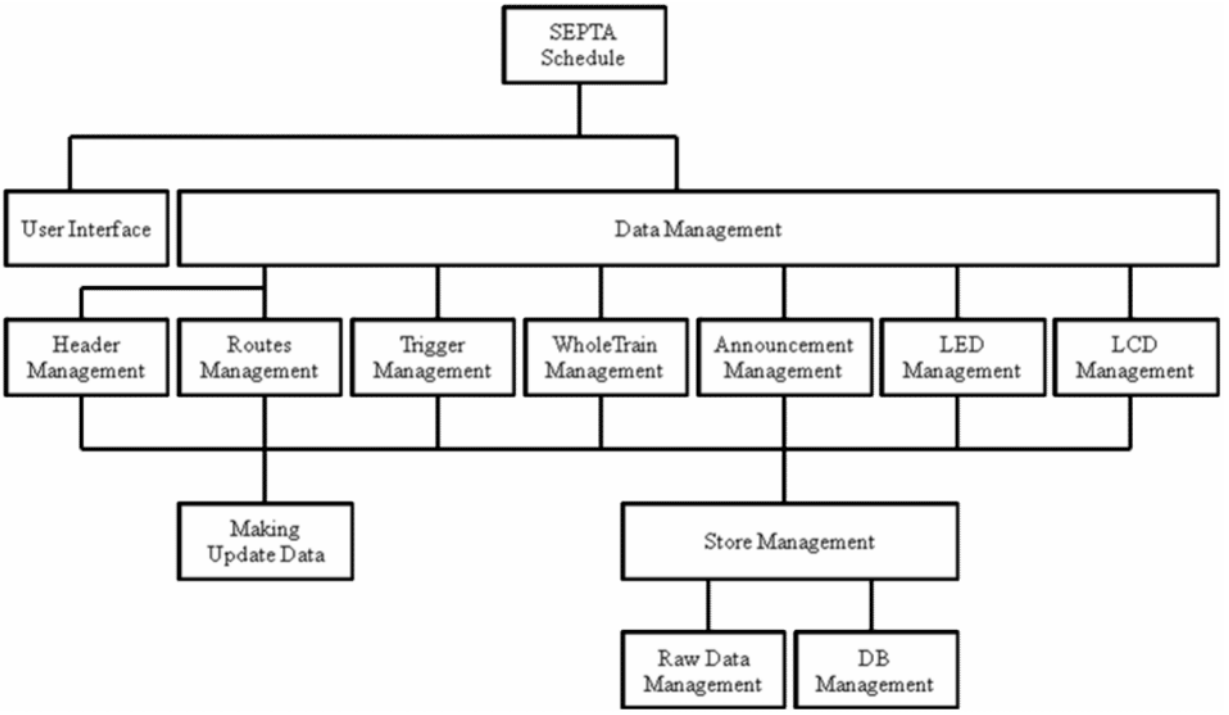


Figure 1. SEPTA Schedule software decomposition diagram

2.1.1. User Interface

Identifier	UserInferface.F
Function	Provides the interface between the user and the SEPTA Schedule.
Sub Entities	

2.1.2. Data Management

Identifier	DataManagement.F
Function	It performs the interface between manager and devices and is able to modify and edit the service data.
Sub Entities	HeaderManagement.F RoutesManagement.F TriggerManagement.F WholeTrainManagement.F AnnouncementManagement.F LEDManagement.F LCDManagement.F

2.1.3. Header Management

Identifier	HeaderManagement.F
Function	Manages the Train Header information that is necessary for the service.
Sub Entities	MakingUpdateData.F StoreManagement.F

2.1.4. Routes Management

Identifier	RoutesManagement.F
Function	Manages the Train Routes information that is necessary for the service.
Sub Entities	MakingUpdateData.F StoreManagement.F

2.1.5. Trigger Management

Identifier	TriggerManagement.F
Function	Manages the Trigger Point information relevant to station.
Sub Entities	MakingUpdateData.F StoreManagement.F

2.1.6. WholeTrain Management

Identifier	WholeTrainManagement.F
Function	Manages the Whole Train announcement information.
Sub Entities	MakingUpdateData.F StoreManagement.F

2.1.7. Announcement Management

Identifier	AnnouncementManagement.F
Function	Manages the Basic announcement and the General announcement information.
Sub Entities	MakingUpdateData.F StoreManagement.F

2.1.8. LED Management

Identifier	LEDManagement.F
Function	Manages the FDI and SDI display information of LED.
Sub Entities	MakingUpdateData.F StoreManagement.F

2.1.9. LCD Management

Identifier	LCDManagement.F
Function	Manages LCD of PP and SP video file information.
Sub Entities	MakingUpdateData.F StoreManagement.F

2.1.10. Making Update Data

Identifier	MakingUpdateData.F
Function	Manages update file information.
Sub Entities	MakingUpdateData.F StoreManagement.F

2.1.11. Store Management

Identifier	StoreManagement.F
Function	Manages operation information and announcement data.
Sub Entities	RawDataManagement.F DBManagement.F

2.1.12. Raw Data Management

Identifier	RawDataManagement.F
Function	Manages the announcement audio file, announcement text and video data.
Sub Entities	

2.1.13. DB Management

Identifier	DBManagement.F
Function	Manages every data of the SEPTA Schedule.
Sub Entities	

2.2. Data Decomposition

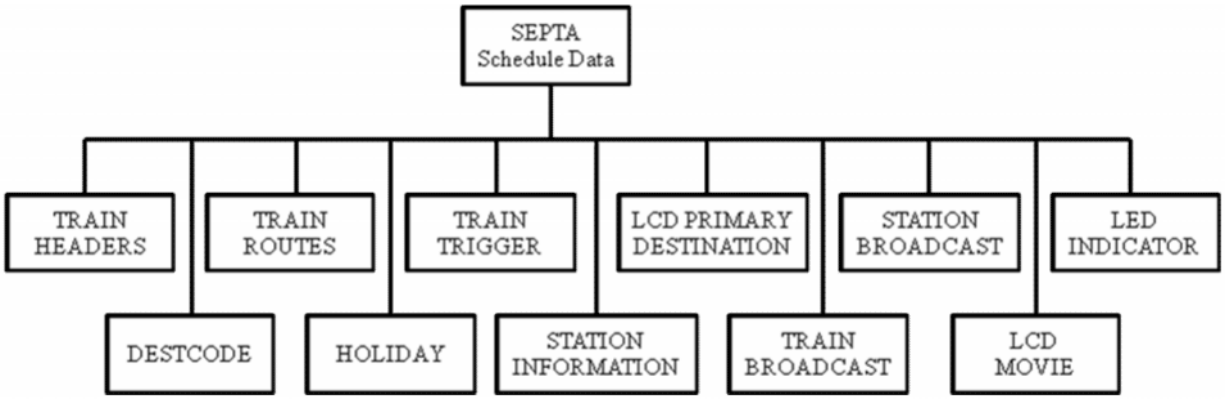


Figure 2. Septa Schedule software data decomposition diagram

2.2.1. TRAIN HEADERS

Identifier	TRAIN_HEADERS.D
Type	DB TABLE
Purpose	It is the table of the train service information.

2.2.2. TRAIN ROUTES

Identifier	TRAIN_ROUTES.D
Type	DB TABLE
Purpose	It is the table of the service route.

2.2.3. STATION INFORMATION

Identifier	STATION_INFORMATION.D
Type	DB TABLE
Purpose	It is the table of station information.

2.2.4. TRAIN BROADCAST

Identifier	TRAIN_BROADCAST.D
Type	DB TABLE
Purpose	It is the table of Whole Train information.

2.2.5. DESTCODE

Identifier	DESTCODE_TABLE.D
Type	DB TABLE
Purpose	It is the table of Destination Code.

2.2.6. LED INDICATOR

Identifier	LED_INDICATOR.D
Type	DB TABLE
Purpose	It is the table of FDI and SDI.

2.2.7. TRAIN TRIGGER

Identifier	TRAIN_TRIGGER.D
Type	DB TABLE
Purpose	It is the table of the GPS coordinate corresponds to station.

2.2.8. LCD MOVIE

Identifier	LCD_MOVIE.D
Type	DB TABLE
Purpose	It is the table of the LCD video list.

2.2.9. HOLIDAY

Identifier	HOLIDAY_TABLE.D
Type	DB TABLE
Purpose	It is the table of holiday operational information.

2.2.10. LCD PRIMARY DESTINATION

Identifier	LCD_PRIMARY_DESTINATION.D
Type	DB TABLE
Purpose	It is the table of the line name that displays on the upper side of the PP.

2.2.11. STATION BROADCAST

Identifier	STATION_BROADCAST.D
Type	DB TABLE
Purpose	It is the table of the LCD Text and the announcement audio file.

3. Dependency Description

This section describes dependency of SEPTA Schedule software design entities using class diagram and data flow diagram.

3.1. Inter-process dependency - Data flow diagram

In this subsection to describes relationship among entities DFD is used.

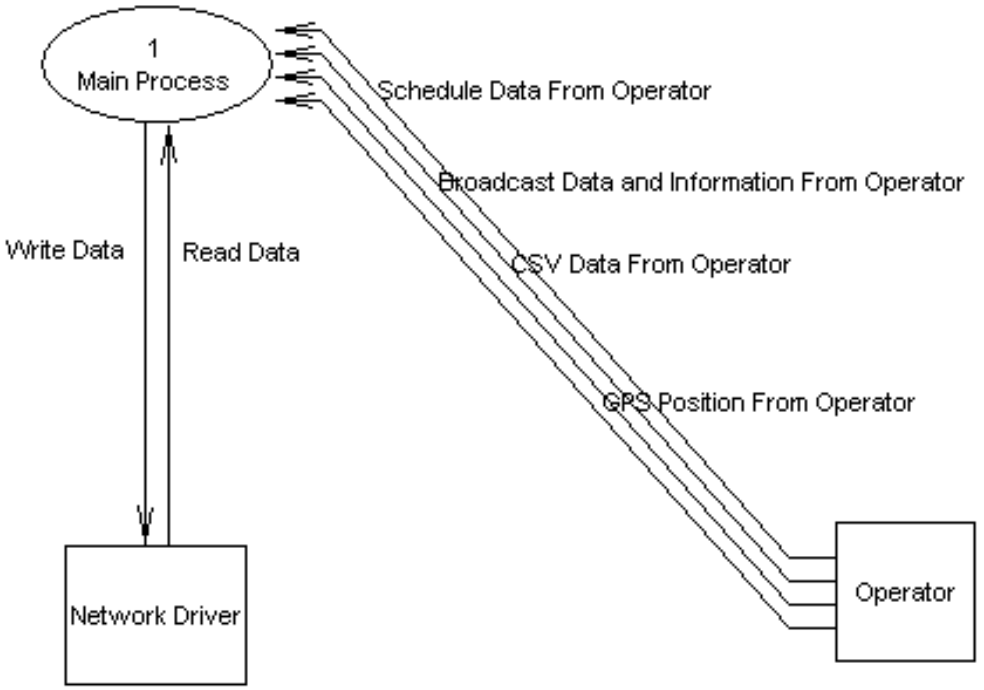


Figure 3. DFD level-0 (Context Diagram)

Identifier	MainProcess.F		
Purpose	It is the process to provide that announcement system is manageable by receiving Schedule, Route, Trigger Point, Holiday and Announcement information of the Operator.		
Source	Operator	Data	Schedule Data From Operator
			Broadcast Data and Information From Operation
			CSV Data From Operator
			GPS Position From Operator
	Network Driver		Read Data
Target	Operator	Data	Update Data From Update Process
	Network Driver		Write Data

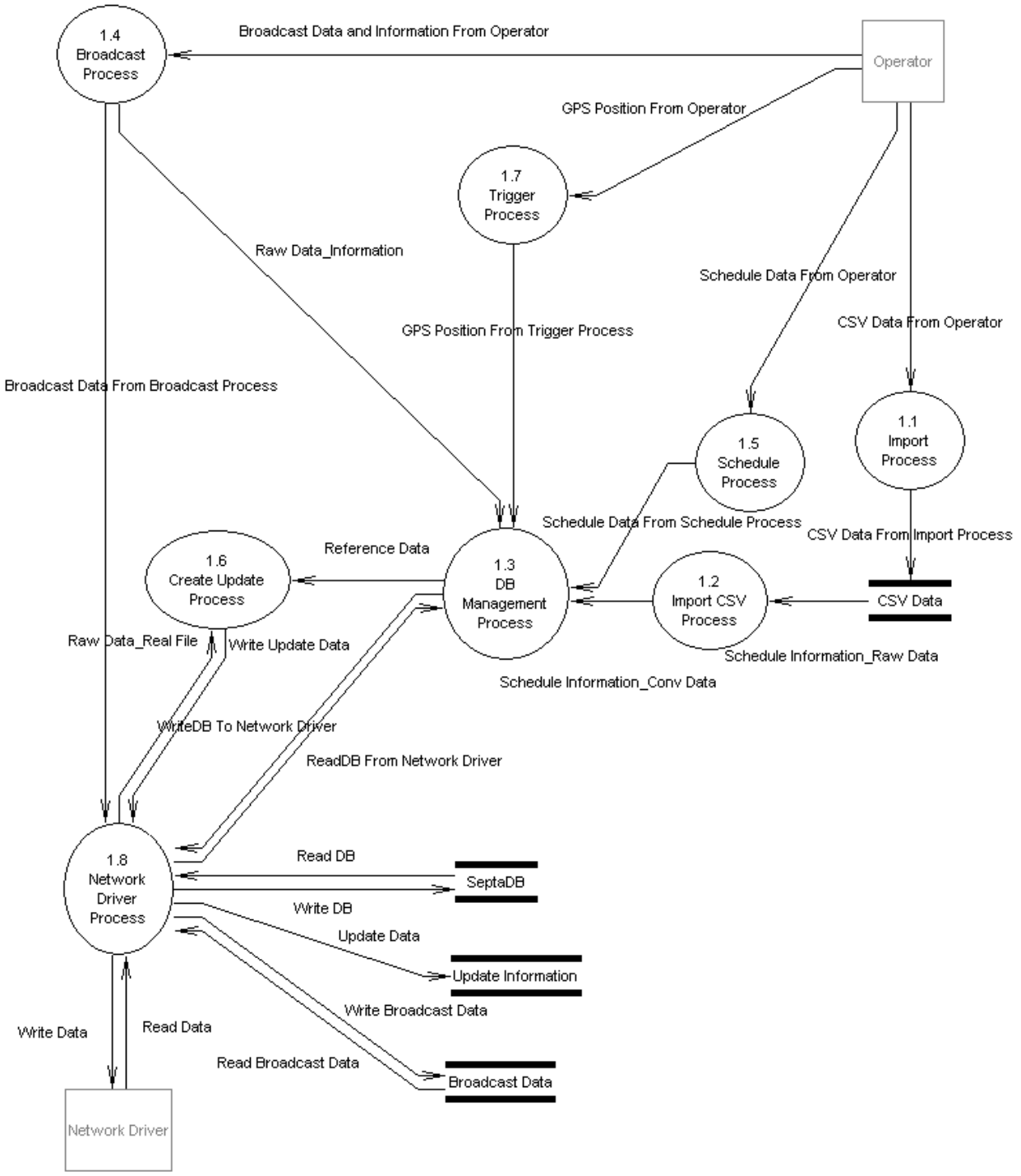


Figure 4. DFD Level-1

Identifier	Import Process		
Purpose	It is the process that creates the management schedule and the route information that the user would enter as the CSV.		
Source	Operator	Data	CSV Data From Operator
Target	CSV Data Store	Data	CSV Data From Import Process

Identifier	Import CSV Process		
Purpose	It is the process that converts created data from the Import Process as the format that is readable on DB.		
Source	Operator	Data	Schedule Information_Raw Data
Target	CSV Data_Store	Data	Schedule Information_Conv Data

Identifier	DB Management Process		
Purpose	It is the process that manages all DB data used in SEPTA Shceduler.		
Source	Import CSV Process	Data	Schedule Information_Conv Data
	Schedule Process		Schedule Data From Schedule Process
	Trigger Process		GPS Position From Trigger Process
	Broadcast Process		Raw Data_Information
	Network Driver Process		ReadDB From Network Driver
Target	Create Update Process	Data	Reference Data
	Network Driver Process		WriteDB To Network Driver
Identifier	Broadcast Process		
Purpose	It is the process for dealing with the LED, LCD, announcement audio file and data.		
Source	Operator	Data	Broadcast Data and Information From Operator
Target	DB Management Process	Data	Raw Data_Information
	Network Driver Process		Broadcast Data From Broadcast Process

Identifier	Schedule Process		
Purpose	It is the process for dealing with the service route, holiday information data.		
Source	Operator	Data	Schedule Data From Operator
Target	DB Management	Data	Schedule Data From Schedule Process

Identifier	Create Update Process		
Purpose	It is process for crating announcement data which is read from DB and Store.		
Source	DB Management Process	Data	Reference Data
	Network Driver Process		Raw Data_Real File
Target	Network Driver Process	Data	Write Update Data

Identifier	Trigger Process		
Purpose	It is the process that is to handle the GPS coordinate.		
Source	Operator	Data	GPS Position From Operator
Target	DB Management	Data	GPS Position From Trigger Process

Identifier	Network Driver Process		
Purpose	It is the process to handle the DB, Raw Data, Update data management through the network.		
Source	Network Driver	Data	Read Data
	SEPTADB_Store		Read DB
	Broadcast Data_Store		Read Broadcast Data
	DB Management Process		WriteDB To Network Driver
	Broadcast Process		Broadcast Data From Broadcast Information
	Create Update Process		Write Update Data

Target	Network Driver	Data	Write Data
	SEPTADB_Store		Write DB
	Broadcast Data_Store		Write Broadcast Data
	Update Information		Update Data
	DB Management Process		ReadDB From Network Driver
	Create Update Process		Raw Data_Real File

3.2. Interporcess dependency - Class Diagram

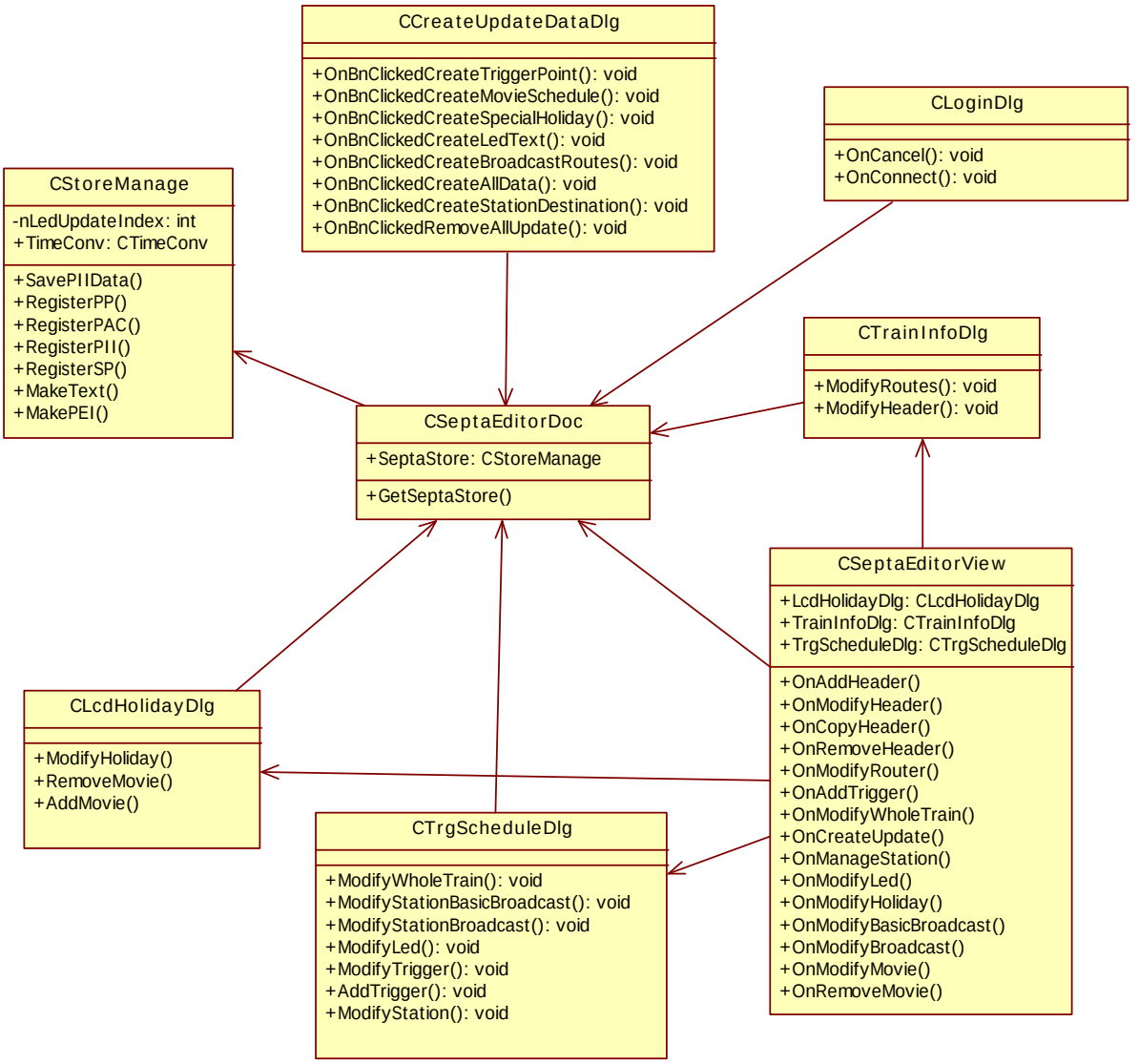


Figure 5. Class Diagram

Class Name	CSeptaEditorView		
Attribute	Data type	Name	Description
	Class	TrainInfoDlg	Manages the Train Header and the Routes information.
	Class	TrgScheduleDlg	Manages the Trigger Point, Station, PAC, LCD and LED information.
	Class	LcdHolidayDlg	Manages the Holiday and the video file information.
Function	Return Type	Name	Description
	void	OnAddHeader	Used when adding the Train Header.
	void	OnModifyHeader	Used when modifying the Train Header.
	void	OnCopyHeader	Used when copying the Train Header.
	void	OnRemoveHeader	Used when deleting the Train Header.
	void	OnModifyRouter	Used when modifying the Train Routes information.
	void	OnModifyTrigger	Used when adding the Trigger Point.
	void	OnModifyWholeTrain	Used when modifying the Whole Train information.
	void	OnCreateUpdate	Used when creating the Update data.
	void	OnManageStation	Used when managing the station information.
	void	OnModifyLed	Used when modifying the LED text.
	void	OnModifyHoliday	Used when modifying the Holiday.
	void	OnModifyBasicBroadcast	Used when modifying the This/Next Announcement.
	void	OnModifyBroadcast	Used when modifying the General announcement.
	void	OnAddMovie	Used when adding the LCD video.
	void	OnRemoveMovie	Used when deleting the LCD video.

Class Name	CLcdHolidayDlg		
Function	Return Type	Name	Description
	void	ModifyHoliday	Used when managing the Holiday.
	void	AddMovie	Used when adding the LCD video.
	void	RemoveMovie	Used when deleting the LCD video.

Class Name	CTrainInfoDlg		
Function	Return Type	Name	Description
	void	ModifyHeader	Used when modifying the Train Header.
	void	ModifyRoutes	Used when modifying the Train Routes.

Class Name	CCreateUpdateDataDlg		
Function	Return Type	Name	Description
	void	OnBnClickedCreateTriggerPoint	Used when creating the Trigger Point.
	void	OnBnClickedCreateMovieSchedule	Used when creating the Movie Schedule.
	void	OnBnClickedCreateSpecialHoliday	Used when creating the Holiday.
	void	OnBnClickedCreateLedText	Used when creating the FDI and SDI text.
	void	OnBnClickedCreateBroadcastRoutes	Used when creating the announcement data and the route.
	void	OnBnClickedCreateAllData	Used when creating all data.
	void	OnBnClickedCreateStationDestination	Used when creating the Station data of the CCP and the Destination.

	void	OnBnClickedRemoveAll Update	Used when deleting temporarily created update data.

Class Name	CStoreManage		
Attribute	Data type	Name	Description
	Int	nLedUpdateIndex	It is used when processing number for Update data.
	Class	TimeConv	It is the class used when changing the Time Format.
Function	Return Type	Name	Description
	void	SavePIIData	Used when creating the LED PII data.
	void	RegisterPP	Used when creating the LCD PP.
	void	RegisterSP	Used when creating the LCD SP Header data.
	void	RegisterPAC	Used when creating the Whole Train, This/Next voice announcement.
	void	MakeText	Used when creating the LCD SP Body data.
	void	MakePEI	Used when creating the PEI Call voice announcement data.

Class Name	CLoginDlg		
Function	Return Type	Name	Description
	void	OnConnect	Used when Network Connect.
	void	OnCancel	Used when Network Cancel.

Class Name	CTrgScheduleDlg		
Function	Return Type	Name	Description
	void	ModifyWholeTrain	Used when modifying the Whole Train information.
	void	ModifyStationBasicBroadcast	Used when modifying the This/Next Announcement.

	void	ModifyStationBroadcast	Used when modifying the General announcement.
	void	ModifyLed	Used when modifying the LED information.
	void	AddTrigger	Used when adding the Trigger information.
	void	ModifyTrigger	Used when modifying Trigger information.
	void	ModifyStation	Used when modifying the Station information.

Class Name	CSeptaEditorDoc		
Function	Return Type	Name	Description
	class	GetSeptaStore	It is used for receiving Instance for SEPTA update.

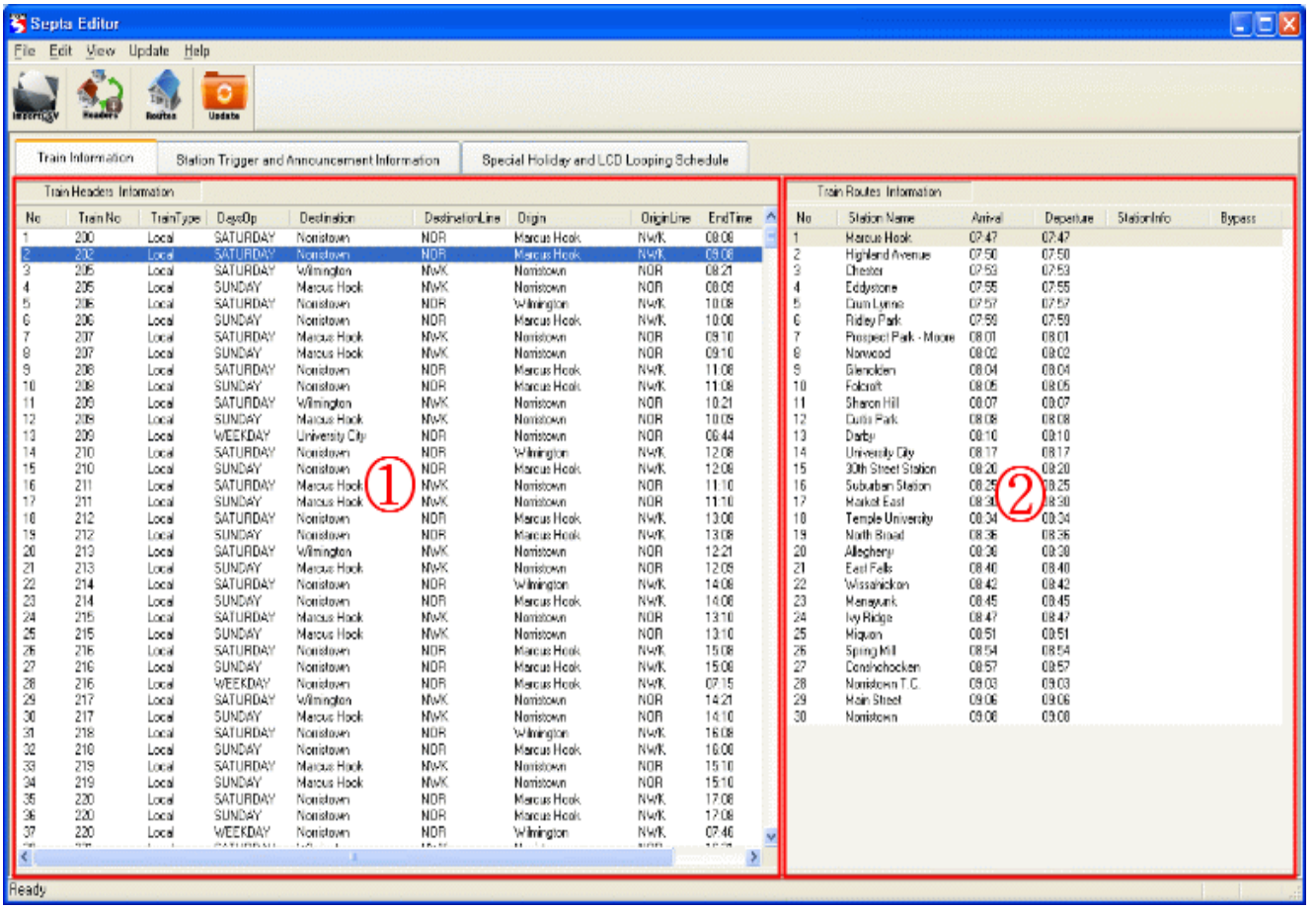
4. Interface Description

4.1. External Interface

4.1.1. User Interface

4.1.1.1. Train Information interface

It is the interface that is able to edit the Headers served on the train and the service routes.

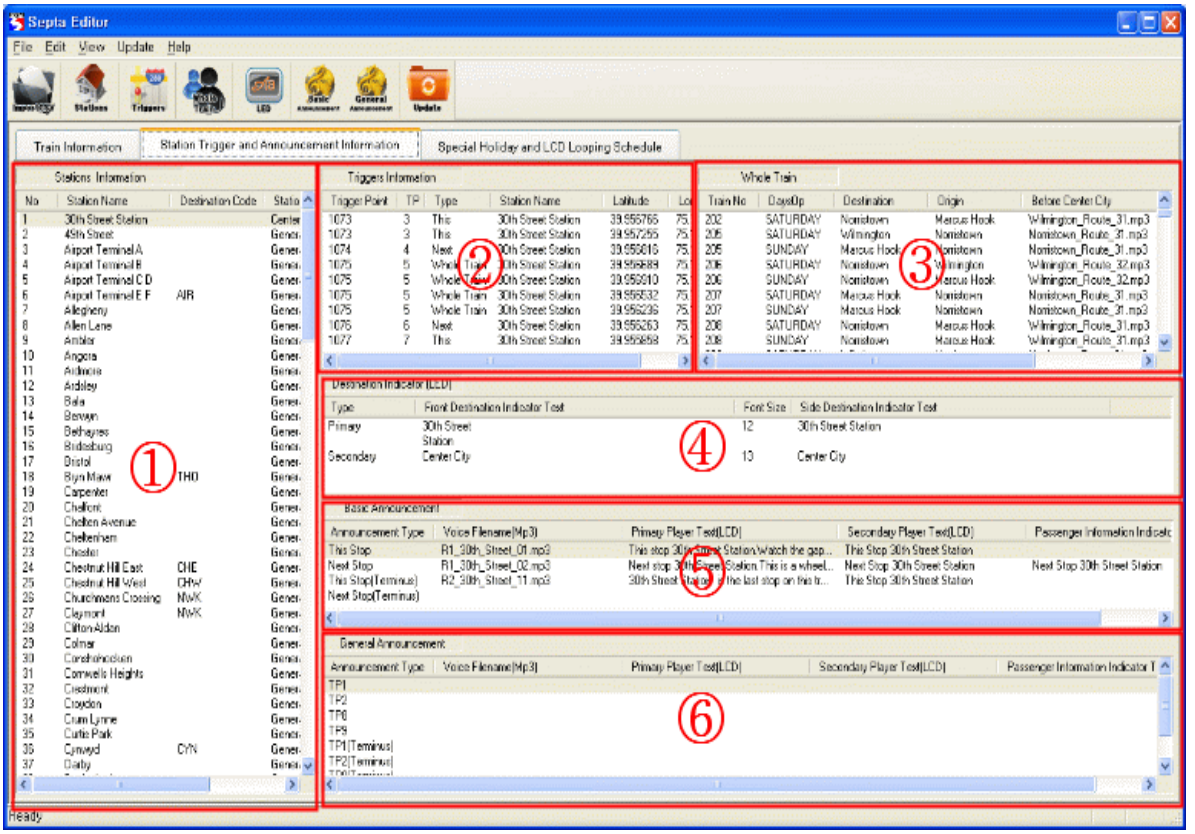


① Train Headers Information: Displays the train service schedule.

② Train Routes Information: Displays the route information relevant to the train service.

4.1.1.2. Station Trigger and Announcement Information

Displays station information, announcement data and Trigger Point information.



① Stations Information: Displays all station names in service.

② Triggers Information: Displays the Trigger Point relevant to station.

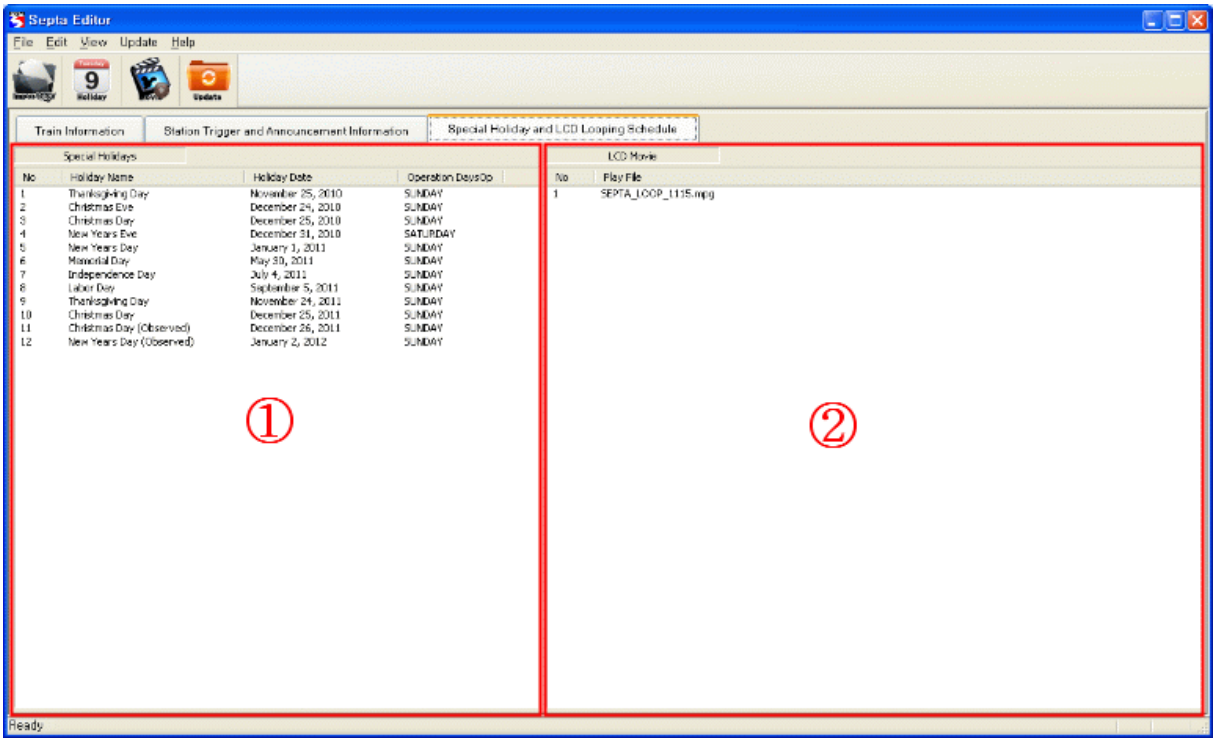
③ Whole Train: Display the Whole Train announcement data when train stop at station.

④ Destination Indicator (LED): Displays the text which will be displayed on LED.

⑤ Basic Announcement: Displays the This/Next/Destination announcement that will be announced before the station or after the station.

⑥ General Announcement : Displays the announcement other than the station announcement.

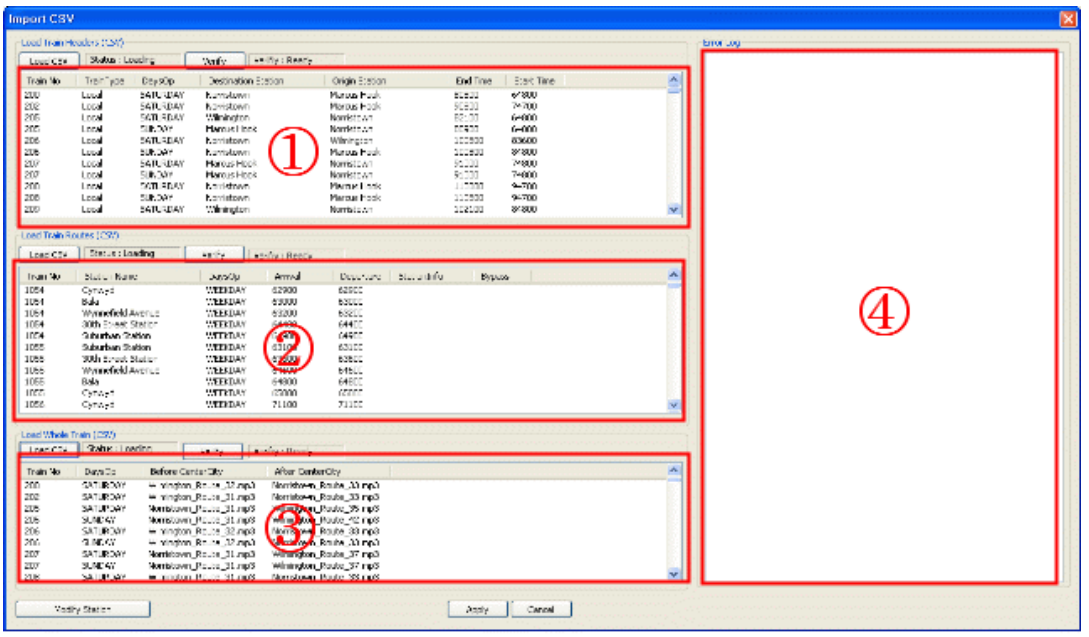
4.1.1.3. Special Holiday and LCD Looping Schedule interface



① Special Holidays: Displays the holiday information that will be designated.

② LCD Movie: Displays the video file list that will be played on the Primary Player.

4.1.1.4. Import CSV interface



Reading the external data is available.

① Load Train Headers: Displays the Train Header CSV contents which have been loaded.

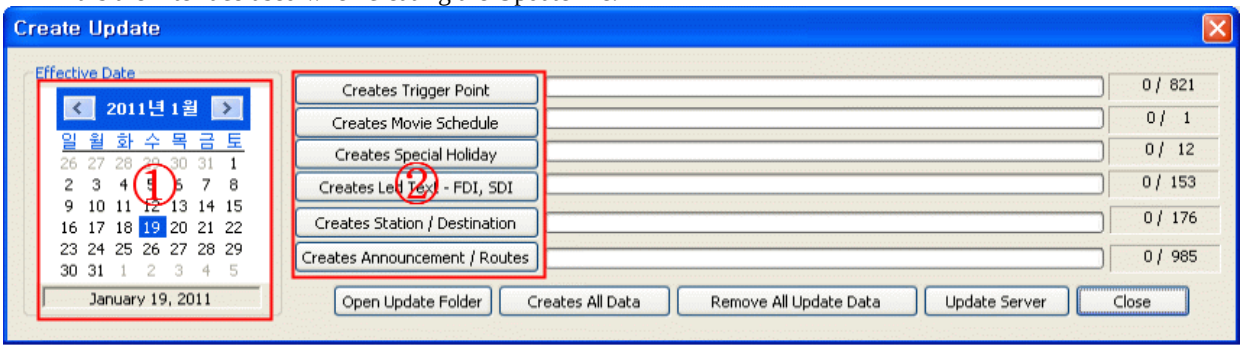
② Load Train Routes: Displays the Train Routes CSV contents which have been loaded.

③ Load Whole Train: Displays the Train Whole Train CSV contents which have been loaded.

④ Error Log : Displays the error contents when loading CSV update and verify data.

4.1.1.5. Create Update Interface

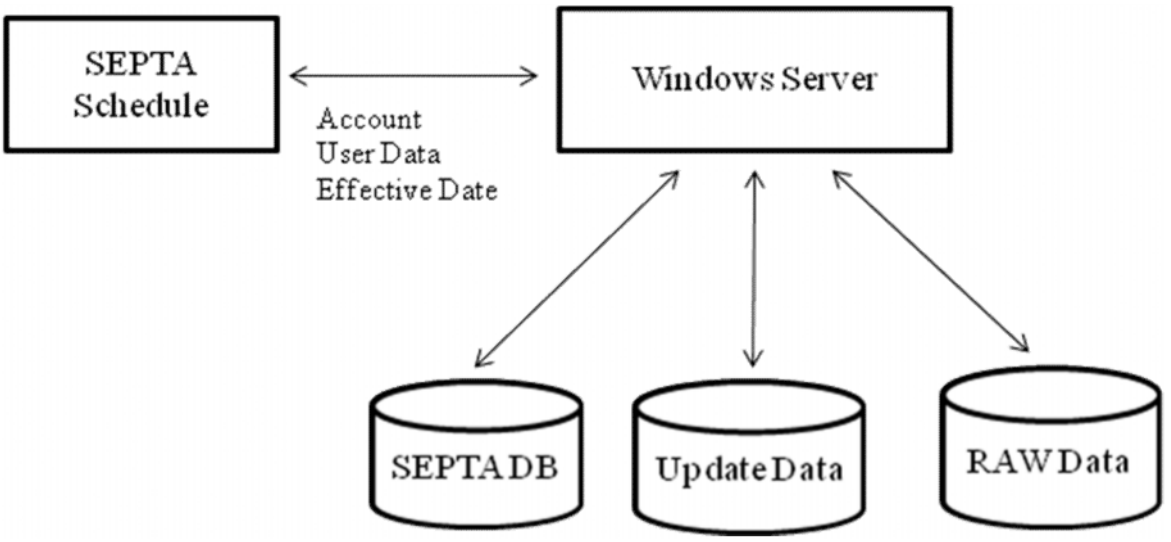
It is the interface used when creating the Update file.



① Effective Date : Selecting the data that will update actually is available.

Creating update file is available.

4.1.2. Network Interface



Windows Network Server is needed.
Manager account and user account shall be assigned.
Using Windows Network Driver when handling data or managing DB is available.

5. Detailed Description

5.1. Module Detailed Design

5.1.1. Add Header Interface

Identifier	SeptaEditorView.OnAddHeader

Processing Logic

LOAD Form FROM TrainHeaderDlg
IF Apply is TRUE THEN
  CALL TrainInfoDlg.WriteSubInfo
  CALL TrainInfoDlg.WriteInfo
  CALL TrgScheduleDlg.WriteWholeTrainInfo
END IF

5.1.2. Modify Header Interface

Identifier	SeptaEditorView.OnModifyHeader

Processing Logic
CALL TrainInfoDlg.ModifyHeader CALL TrgScheduleDlg.WriteWholeTrainInfo

5.1.3. Copy Header Interface

Identifier	SeptaEditorView.OnCopyHeader

Processing Logic
CALL TrainInfoDlg.CopyHeader

5.1.4. Remove Header Interface

Identifier	SeptaEditorView.OnRemoveHeader

Processing Logic

IF SelectedHeader is TRUE THEN
  DELETE SelectedData FROM TRAIN_HEADERS DB
  DELETE SelectedData FROM FROM TRAIN_BROADCAST DB
  CALL TrainInfoDlg.WriteSubInfo
END IF

5.1.5. Modify Router Interface

Identifier	SeptaEditorView.OnModifyRouter
Processing Logic	
CALL TrainInfoDlg.ModifyRoutes	

5.1.6. Add Trigger Interface

Identifier	SeptaEditorView.OnAddTrigger
Processing Logic	
CALL TrgScheduleDlg.AddTrigger	

5.1.7. Modify Whole Train Interface

Identifier	SeptaEditorView.OnModifyWholeTrain
Processing Logic	
CALL TrgScheduleDlg.ModifyWholeTrain	

5.1.8. Create Update Interface

Identifier	SeptaEditorView.OnCreateUpdate
Processing Logic	
LOAD Form FROM CUpdateDataDlg	

5.1.9. Manage Station Interface

Identifier	SeptaEditorView.OnManageStation
Processing Logic	
CALL TrgScheduleDlg.ModifyStation	

5.1.10. Modify Led Interface

Identifier	SeptaEditorView.OnModifyLed
Processing Logic	
CALL TrgScheduleDlg.ModifyLed	

5.1.11. Modify Holiday Interface

Identifier	SeptaEditorView.OnModifyHoliday
Processing Logic	
CALL LcdHolidayDlg.ModifyHoliday	

5.1.12. Modify Basic Broadcast Interface

Identifier	SeptaEditorView.OnModifyBasicBroadcast
Processing Logic	
CALL TrgScheduleDlg.ModifyStationBasicBroadcast	

5.1.13. Modify General Broadcast Interface

Identifier	SeptaEditorView.OnModifyBroadcast
Processing Logic	
CALL TrgScheduleDlg.ModifyStationBroadcast	

5.1.14. Add Movie Interface

Identifier	SeptaEditorView.OnAddMovie
Processing Logic	
CALL LcdHolidayDlg.AddMovie	

5.1.15. Remove Movie Interface

Identifier	SeptaEditorView.OnRemoveMovie
Processing Logic	
CALL LcdHolidayDlg.RemoveMovie	

5.1.16. Modify Holiday

Identifier	CLcdHolidayDlg.ModifyHoliday
Processing Logic	
LOAD Form FROM CSpecialHolidayDlg IF APPLY is TRUE CALL WriteHoliday END IF	

5.1.17. Add Movie

Identifier	CLcdHolidayDlg.AddMovie
Processing Logic	

READ MovieFile
IF MovieFile is exist THEN
  WRITE MovieFile to Disk
  WRITE MovieFile to LCD_MOVIE DB
END IF	

5.1.18. Remove Movie

Identifier	CLcdHolidayDlg.RemoveMovie

Processing Logic

IF SelectedMovie is TRUE THEN
DELETE MovieFile FROM LCD_MOVIE
DELETE MovieFile FROM DISK
END IF

5.1.19. Modify Header

Identifier	CTrainInfoDlg.ModifyHeader

Processing Logic

LOAD Form FROM CTrainHeaderDlg
IF APPLY is TRUE THEN
CALL WriteInfo
CALL WriteSubInfo
END IF

5.1.20. Modify Route

Identifier	CTrainInfoDlg.ModifyRoutes

Processing Logic

LOAD Form FROM TrainRouteDlg
IF APPLY is TRUE THEN
CALL WriteSubInfo
END IF

5.1.21. Creates Trigger Point

Identifier	CCreateUpdateDataDlg.OnBnClickedCreateTriggerPoint

Processing Logic
READ TriggerPoint FROM TRAIN_TRIGGER WRITE UpdateTriggerData TO Disk

5.1.22. Creates Movie Schedule

Identifier	CCreateUpdateDataDlg.OnBnClickedCreateMovieSchedule

Processing Logic
READ MovieSchedule FROM LCD_MOVIE DB WRITE UpdateMovieSchedule TO Disk WRITE UpdateMovieData TO Disk

5.1.23. Creates Special Holiday

Identifier	CCreateUpdateDataDlg.OnBnClickedCreateSpecialHoliday
Processing Logic	
READ HolidaySchedule FROM HOLIDAY_TABLE DB WRITE UpdateHolidaySchedule TO DISK	

5.1.24. Creates Led Text

Identifier	CCreateUpdateDataDlg.OnBnClickedCreateLedText
Processing Logic	

READ DestinationStation FROM LED_INDICATOR DB
WHILE (DestinationStation is not EOF)
Call CSpetaEditorDoc.MakeFDI
Call CSpetaEditorDoc.MakeSDI
Call CSpetaEditorDoc.WriteVector
END WHILE	

5.1.25. Creates Broadcast and Routes

Identifier	CCreateUpdateDataDlg.OnBnClickedCreateBroadcastRoutes
Processing Logic	

READ DestinationText FROM PRIMARY_DESTINATION DB
READ TrainHeader FROM TRAIN_HEADERS DB
READ TrainBroadcast FROM TRAIN_ROUTES DB
WHILE TrainHeader is not EOF
READ TrainRoutes FROM TRAIN_ROUTES DB
WHILE TrainRoutes is not EOF
READ StationBroadcast FROM STATION_BROADCAST DB
WHILE StationBroadcast is not EOF
IF Voice File is exist THEN
Call CSpetaEditorDoc.RegisterPAC
Call CSpetaEditorDoc.RegisterPII
Call CSpetaEditorDoc.RegisterSP
WRITE Update Information TO DISK
END IF
END WHILE
IF WholeTrain is exist THEN
WRITE UpdateWholeTrain TO DISK
END IF
END WHILE
END WHILE
WRITE CSpetaEditorDoc.SavePIIData	

5.1.26. Creates All Data

Identifier	CCreateUpdateDataDlg.OnBnClickedCreateAllData

Processing Logic
CALL OnBnClickedCreateTriggerPoint CALL OnBnClickedCreateMovieSchedule CALL OnBnClickedCreateSpecialHoliday CALL OnBnClickedCreateLedText CALL OnBnClickedCreateStationDestination CALL OnBnClickedCreateBroadcastRoutes

5.1.27. Creates Station and Destination

Identifier	CCreateUpdateDataDlg.OnBnClickedCreateStationDestination
Processing Logic	

READ StationInformation FROM STATION_INFORMATION DB
READ DestinationInformation FROM DESTCODE_TABLE DB
WHILE (StationInformation is not EOF)
  Write UpdateStationInformaion TO DISK
END WHILE
WHILE (DestinationInformation is not EOF)
  Write UpdateDestinationInformaion TO DISK
END WHILE	

5.1.28. Remove All Update

Identifier	CCreateUpdateDataDlg.OnBnClickedRemoveAllUpdate
Processing Logic	
DELETE CSpetaEditorDoc.GetUpdatePath() FROM DISK CALL CSpetaEditorDoc.SetDate(EffectiveDate)	

5.1.29. Creates All Data

Identifier	CCreateUpdateDataDlg.OnBnClickedCreateAllData
Processing Logic	
CALL OnBnClickedCreateTriggerPoint CALL OnBnClickedCreateMovieSchedule CALL OnBnClickedCreateSpecialHoliday CALL OnBnClickedCreateLedText CALL OnBnClickedCreateStationDestination CALL OnBnClickedCreateBroadcastRoutes	

5.1.30. Save Passenger Indicator Information

Identifier	CStoreManage.SavePIIData
Processing Logic	

READ PIIVector FROM Memory
WHILE (PIIVector is not EOF)
  WRITE UpdatePIIVector TO DISK
END IF	

5.1.31. Register Primary Player Information

Identifier	CStoreManage.RegisterPP
Processing Logic	
READ PPText FROM DISK WRITE PPHeader TO DISK WRITE PPBody TO DISK	

5.1.32. Register Secondary Player Information

Identifier	CStoreManage.RegisterSP
Processing Logic	
READ SPText FROM DISK WRITE UpdateSPText TO DISK	

5.1.33. Register PAC Information

Identifier	CStoreManage.RegisterPAC
Processing Logic	
READ PAC File FROM DISK WRITE UpdatePAC File TO DISK	

5.1.34. Register PII Information

Identifier	CStoreManage.RegisterPII
Processing Logic	
WRITE PIIText TO MEMORY	

5.1.35. Make Primary Text

Identifier	CStoreManage.MakeText
Processing Logic	
WRITE PPText FROM DISK WRITE UpdatePPText TO DISK	

5.1.36. Make PEI Call

Identifier	CStoreManage.MakePEI
Processing Logic	
READ PEICALL FROM DISK WHILE PEICALL is exist WRITE UpdatePEICALL TO DISK END WHILE	

5.1.37. Login Network Server

Identifier	CLoginDlg.OnConnect
Processing Logic	

IF NetworkDriver is TRUE THEN
  PERFORM NetworkConnect
  IF Connect is TRUE THEN
  PERFORM NETWORK_TYPE
  END IF
ELSE
  PERFORM LOCAL _TYPE
END IF	

5.1.38. Cancel Login

Identifier	CLoginDlg.OnCancel
Processing Logic	
EXIT Program	

5.1.39. Modify Whole Train

Identifier	CTrgScheduleDlg.ModifyWholeTrain
Processing Logic	

LOAD Form FROM CWholeTrain
IF APPLY is TRUE THEN
  CALL WriteWholeTrainInfo
END IF	

5.1.40. Modify Station Basic Broadcast

Identifier	CTrgScheduleDlg.ModifyStationBasicBroadcast
Processing Logic	

LOAD Form FROM CStationBroadcast (Basic)
IF APPLY is TRUE THEN
  CALL WriteBroadcast
END IF	

5.1.41. Modify Station Broadcast

Identifier	CTrgScheduleDlg.ModifyStationBroadcast
Processing Logic	

LOAD Form FROM CStationBroadcast (General)
IF APPLY is TRUE THEN
  CALL WriteBroadcast
END IF	

5.1.42. Modify Led Data

Identifier	CTrgScheduleDlg.ModifyLed
Processing Logic	

LOAD Form FROM CLedDlg
IF APPLY is TRUE THEN
  CALL WriteLedIndicator
END IF	

5.1.43. Add Trigger Point

Identifier	
Processing Logic	

LOAD Form FROM CAppTriggerPointDlg
IF APPLY is TRUE THEN
  CALL WriteTrigger
END IF	

5.1.44. Modify Trigger Point

Identifier	CTrgScheduleDlg.ModifyTrigger
Processing Logic	

LOAD Form FROM CTriggerPointDlg
IF APPLY is TRUE THEN
  CALL WriteTrigger
END IF	

5.1.45. Modify Station

Identifier	CTrgScheduleDlg.ModifyStation
Processing Logic	
RETURN SeptaStore	

5.2. Data Detailed Design

5.2.1. ERD

Use the Database of Figure 2.

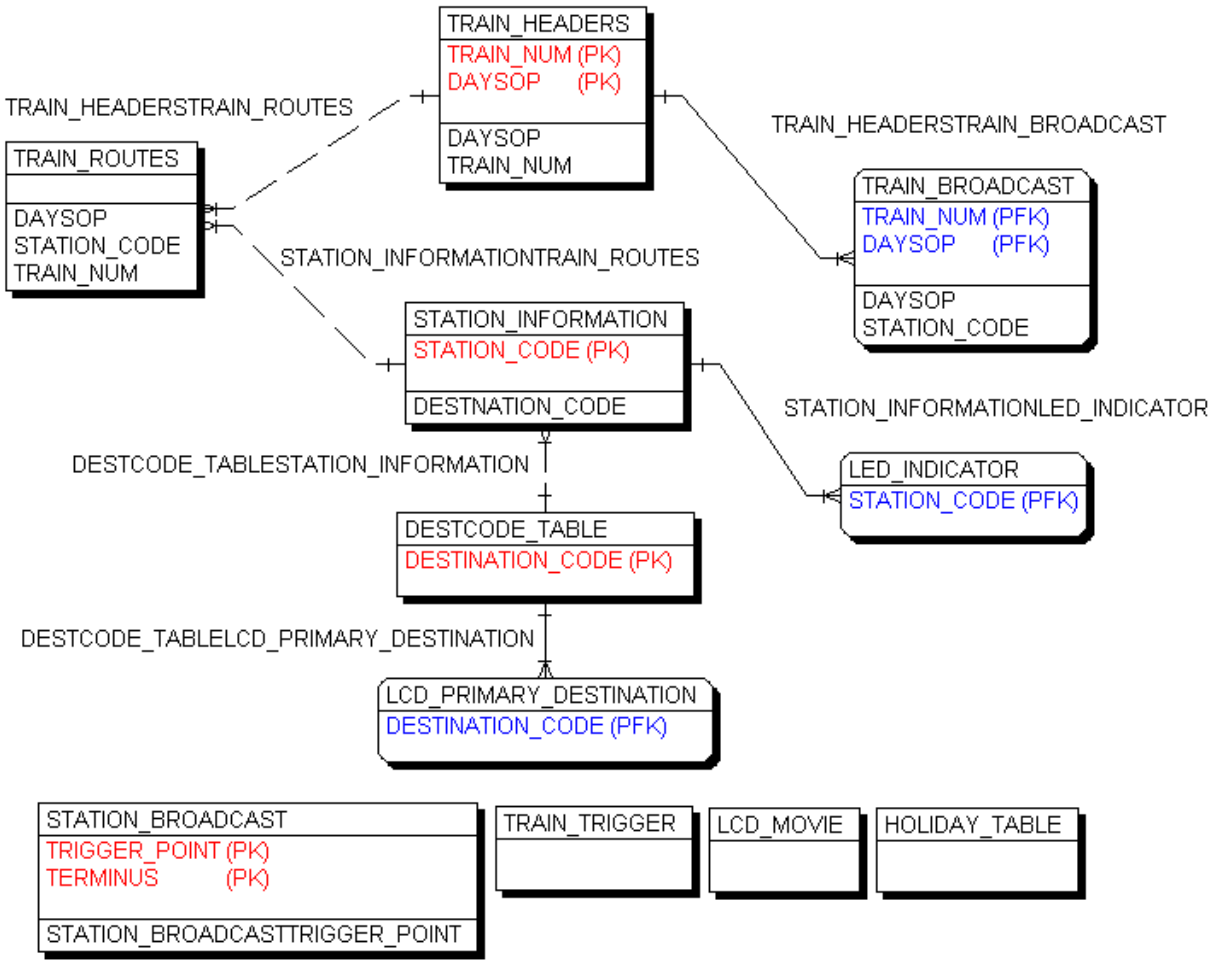


Figure 6. SEPTA Scheduler Software ERD

5.2.2. TRAIN Headers

Table Name	TRAIN_HEADERS			
Column name	Key	Data Type	Relationships	Description
TRAIN_NUM	PK	Integer	TRAIN_ROUTES TRAIN_BROADCAST	Service Train Number
DAYSOP	PK	Text(255)		Service Day
TRAIN_TYPE		Text(255)		Service Type Status
DESTINATION		Text(255)		Service Final station Code
ORIGIN		Text(255)		Service Origin station Code
END_TIME		Integer		Service ending time
START_TIME		Integer		Service starting time

5.2.3. Train Routes

Table Name	TRAIN_ROUTES			
Column name	Key	Data Type	Relationships	Description
TRAIN_NUM	FK	Integer	TRAIN_HEADERS	Train number relevant to the routes.
DAYSOP	FK	Text(255)		Day relevant to the routes.
STATION_CODE	FK	Text(255)	STATION_INFORMATION	Station Code
ARRIVAL_TIME		Integer		Arrival Time at the station
DEPARTURE_TIME		Integer		Departure Time at the Station
STATION_INFO		Text(255)		Station Type Status
BYPASS		Text(255)		SKIP status of the station
SUB_INDEX		Integer		Index of the station code

5.2.4. Station Information

Table Name	STATION_INFORMATION			
Column name	Key	Data Type	Relationships	Description
STATION_CODE	PK	Text(255)	TRAIN_ROUTES	Station Code
CENTER_CITY		Integer		Station Type
STATION_NAME		Text(255)		Station Name
DESTINATION_CODE		Integer		Line Name Code
TERMINUS		Boolean		Destination Status
TRIGGER_ORDER		Integer		Trigger Point Direction
CCP_EXPAND_NAME		Text(255)		CCP Display station name
ALTERNATE		Boolean		Alternate Status
SECONDARY_STATION_NAME		Text(255)		Alternate station name

5.2.5. Train Broadcast

Table Name	TRAIN_BROADCAST.D			
Column name	Key	Data Type	Relationships	Description
TRAIN_NUM	FK	Integer	TRAIN_HEADERS	It is the Train Number which whole train will be announced.
DAYSOP	FK	Text(255)		It is the day which whole train will be announced.
BEFORE_WHOLE_TRAIN		Text(255)		It is the file name that will be announced before the Center City.
AFTER_WHOLE_TRAIN		Text(255)		It is the file name that will be announced after the Center City.

5.2.6. Destination Code

Table Name	DESTCODE_TABLE.D			
Column name	Key	Data Type	Relationships	Description
TRAIN_NUM	FK	Integer	TRAIN_HEADERS	It is the Train Number which whole train will be announced.
DAYSOP	FK	Text(255)		It is the day which whole train will be announced.
BEFORE_WHOLE_TRAIN		Text(255)		It is the file name that will be announced before the Center City.
AFTER_WHOLE_TRAIN		Text(255)		It is the file name that will be announced after the Center City.

5.2.7. Led Indicator

Table Name	LED_INDICATOR.D			
Column name	Key	Data Type	Relationships	Description
STATION_CODE	FK	Text(255)	STATION_INFORMATION	Code name of LED
FRONT_INDICATOR_TEXT		Text(255)		The first text which will be displayed on the FDI when using the basic station name.
FRONT_INDICATOR_TEXT1		Text(255)		The second text which will be displayed on the FDI when using the basic station name.
SECONDARY_FRONT_INDICATOR_TEXT		Text(255)		The first text which will be displayed on the FDI when using the ALTERNATE.
SECONDARY_FRONT_INDICATOR_TEXT1		Text(255)		The second text which will be displayed on the FDI when using the ALTERNATE.
SIDE_INDICATOR_TEXT		Text(255)		The text which will be displayed on the SDI when using the basic station name.
SECONDARY_SIDE		Text(255)		The text which will be displayed on the SDI

_INDICATOR_TEXT				when using the ALTERNATE.
FRONT_INDICATOR_ALIGN		Integer		Array status of the FDI
SIDE_INDICATOR_ALIGN		Integer		Array status of the SDI
FRONT_INDICATOR_TEXTSIZE		Integer		FDI Text size data when using the basic station name.
SECONDARY_FRONT_INDICATOR_TEXTSIZE		Integer		FDI Text size data when using the ALTERNATE

5.2.8. Trigger

Table Name	TRAIN_TRIGGER.D			
Column name	Key	Data Type	Relationships	Description
TRIGGER_POINT		Long		Trigger Point Code
LONGITUDE		Text(255)		Longitude relevant to the Trigger Point
LATITUDE		Text(255)		Latitude relevant to the Trigger Point
INDEX		Integer		Index of the Trigger Point

5.2.9. Movie

Table Name	LCD_MOVIE.D			
Column name	Key	Data Type	Relationships	Description
INDEX		Long		Video file Index
PLAY_FILE		Text(255)		Video File Name

5.2.10. Holiday

Table Name	HOLIDAY_TABLE.D			
Column name	Key	Data Type	Relationships	Description
HOLIDAY_YEAR		Long		Year of the Holiday
HOLIDAY_MONTH		Long		Month of the Holiday
HOLIDAY_DAY		Long		Day of the Holiday
HOLIDAY_WEEKS		Long		Week of the Holiday
HOLIDAY_DAPSOP		Long		Ordinal of Holiday
HOLIDAY_NAME		Text(255)		Name of the Holiday
HOLIDAY_TYPE		Boolean		Type Status of the Holiday
HOLIDAY_OPERATION_DAYSOP		Long		Designating the day schedule that will be served on the holiday

5.2.11. LCD Primary Header

Table Name	LCD_PRIMARY_DESTINATION.D			
Column name	Key	Data Type	Relationships	Description
DESTINATION_CODE	FK	Long		Line Code
HEADER_TEXT		Text(50)		Text of the SP Header
FONT_SIZE		Long		Text Size of the SP Header
FONT_COLOR		Long		Text Color of the SP Header

5.2.12. Station Broadcast

Table Name	STATION_BROADCAST.D			
Column name	Key	Data Type	Relationships	Description
TRIGGER_POINT	PK	Text(255)		Trigger Point Code
TERMINUS	PK	Boolean		Destination Status of the Trigger Point
VOICE_FILE		Text(255)		The MP3 file that is to be announced.
PRIMARY_PLAYER_DELAYTIME		Integer		Display time of the PP text
PRIMARY_PLAYER_TIME_DELAYTIME		Integer		Display time of the clock on the PP
PRIMARY_PLAYER_TEXT		Text(255)		Display text on the PP body part.
PRIMARY_PLAYER_FONTSIZE		Integer		Font Size of the PP body part
PRIMARY_PLAYER_FONTCOLOR		Integer		Font Color of the PP Body part
SECONDARY_PLAYER_TEXT		Text(255)		Text that will be displayed on the SP
PASSENGER_INFO_INDICATOR_TEXT		Text(255)		Text that will be displayed on the PII